



Power Platform Capability/Skillset Overview

Fully Functional Business Solutions

David Clerk

Owner

Revised: 202509

Agenda

- Introduction
- Execution Strategy
- - Project Overview
- - Requirements Analysis
- - Project Deliverables
- - Models
- - Conclusion



Introduction

David Clerk - Fully Functional Business Solutions (FullyFBS)

David Clerk is the founder of Fully Functional Business Solutions (FullyFBS), a start-up business which develops and maintains information technology solutions for complex business problems. I am a diverse problem solver with extensive training, knowledge and experience in data analytics via a variety of industries. I am highly proficient in the application of statistical techniques, process improvement methodologies, database management, qualitative and quantitative data analytics, visualizations, and reporting. My technical acumen, interactive leadership and proven ability to deliver great solutions provides consistent and reliable positive impacts to businesses and strategic objectives.



My approach aims to:

- Employ a strategy that combines traditional planning rigor with Agile principles to adapt to the nature of work: some parts are predictable and require structured execution, while others need iterative, collaborative problem-solving.
- Retain flexibility, responsiveness, and a pragmatic approach to project delivery rather than strictly adhering to a single methodology.



Use Case (Fictional)

CS Inc. provides maintenance services for material handling and transportation equipment. CS Inc. has a customized CRM which it uses to manage its inventory, customers, and services. The current CRM has undergone multiple customizations and upgrades and has become very costly to implement new processes that match the ever-growing business needs. Over the past two years CS Inc. has experienced a 15% increase in services outsourced to sub-contractors; however, the current system does not have automated processes that allow the company to manage sub-contractors, sub-contractor servicing, and the sub-contractor invoicing and payments process. CS Inc. has identified a high priority need for a process to manage sub-contractors. The executive board of CS needs to make a well-informed decision which will determine if CS Inc. needs to purchase a new CRM or build business solutions from its existing platforms. CS Inc.'s needs an operational model of a scalable multi-platform automated solution which it can integrate with its current CRM to help management determine if the Power Platform will meet its requirements.

This presentation demonstrates:

- Seamless integration of Power Platform with Azure resources
- My inherent and retained expertise in cloud-based and cross-platform data integration, enabling consistent delivery of high-value solutions

Execution Strategy

Strategy

FullyFBS execution strategy is built on Hybrid Agile approach. FullyFBS 's standard execution strategy integrates an incremental and iterative development process which leverages the [Agile Alliance manifesto](#). FullyFBS will incessantly advocate:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

Deliverables

With the understanding this project is for academic purpose and within the constraints of limited time, rendering of the following deliverables with deem this project a success:

- **Data Sources with ERD** -Used DataVerse and Azure SQL)
- **Customized and OTB Apps** - Use of Azure DB, PowerFX, Power Flows, and SQL Store Procedures
- **Visualizations and Dashboards** - PowerBi Visuals with Use of Python Mapplot ect...
- **Web UI** - UI hosted by Portal Management with embedded Azure AD, PowerApps, and PowerBI
- **Production of Invoices** – Used Power Bi Report Builder for paginated reports

Project Overview

Opportunity

FullyFBS intends to capitalize on the plethora of the training opportunities this project provides. FullyFBS intends to provide CS Inc. with a fully function business solution by developing an application (as a model) for CS to use as a technician invoice and payments application (TIPA). Although the Use Case is fictional; the intent of this project is to provide a proof of concept for using the Power Platform for the following:

- **Data and Connections:** Employ SQL/Azure DB, DataVerse and various connectors to create, manage, store and query data.
- **Analytics and Visualizations:** Utilize qualitative and quantitative metrics, KPIs, visualizations, and Dashboards via Power Bi/Python.
- **Web Portal:** Harness the capability of creating a Web Portal to broadcast applications, visualizations, and manipulate data.
- **Application Development:** Design and develop an application via Power Apps. The platform will provide tools and features for data input, output, dataflows, workflows and automation.

-

Requirements Summary

Proposed Solution

- An invoice and payment application built on a single platform which will allow connection to existing platforms, applications and/or services from on-premises or mobile devices.
- Low/No Code application platform supported by easy to learn features, scalability, cost saving maintenance and upgrade cost.
- Method for ranking service opportunities and technicians.
- Method for ranking service opportunities and technicians, grouping service requests into maintenance groups, predicting customer maintenance needs, and assigning the service request to preferred technicians based on availability.
- A scalable interactive knowledge base that provides best answers from the user's perspective/category.
- Visualizations, dashboards and data insights on customers, technicians, invoices, and payments led by business rules, KPIs, and metrics.
- Method for determining, approving, and overriding service completions.

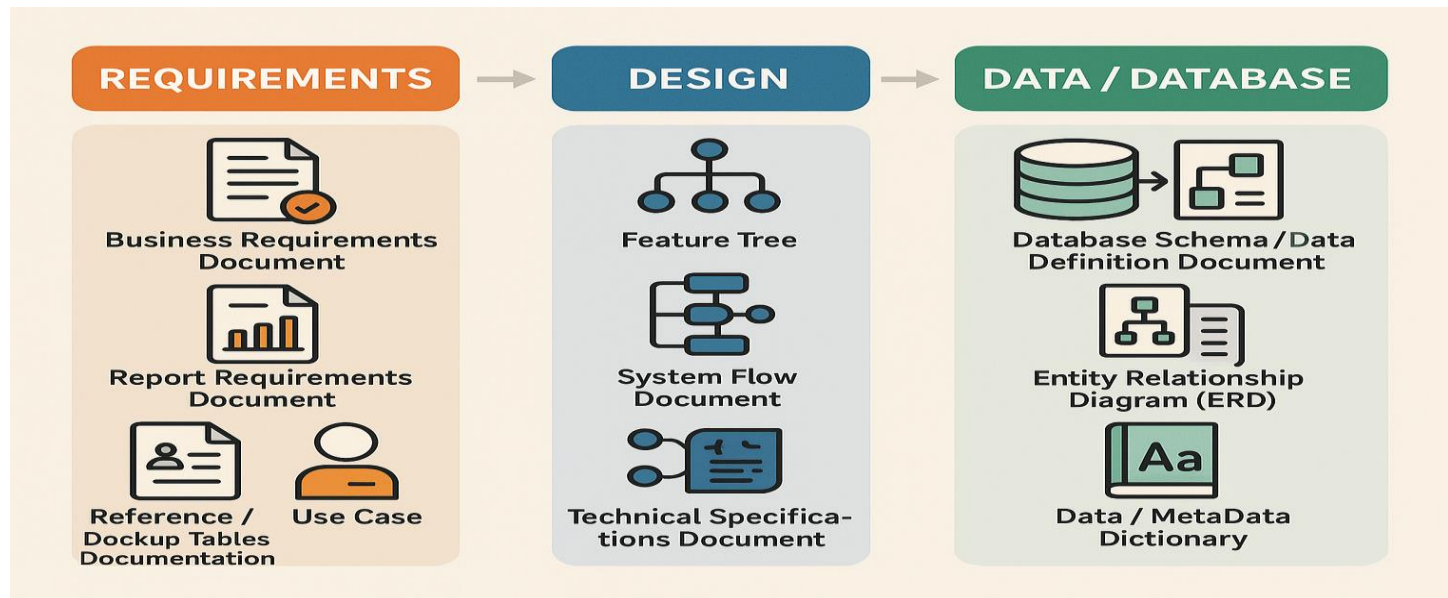
Software Development Life Cycle (SDLC)

SDLC

- SDLC is the overall process of planning, developing, testing, deploying, and maintaining software. Different artifacts or documents are created during SDLC phases to ensure quality, clarity, and alignment with requirements.

Artifacts

- In layman's terms SDLC artifacts are planning and other technical documents. However, the time and effort that goes into creating, maintain and proper disposition of these documents gain them the term artifacts or works of art. Due to time constraints and the fictionality of this project, the graphic below depicts my preferred employment of each SDLC artifact.





Project Management

Excel CapStoneRequirementsWorksheet A^R - Saved

Search (Alt + Q)

File Home Insert Draw Page Layout Formulas Data Review View Help Editing

Undo Paste Cut Copy Format Painter Clipboard Font Alignment Number Styles Cells Editing

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Description	Start Date	End Date	Duration	Status	KLO												
2	Project Start - TIPA Capstone	1/13/2022	3/10/2022	56		Data Science Immersive												
3	Phase 1 - Technical Feasibility Assessment	1/13/2022	2/4/2022	22		SQL, DataVerse, Excel, MS Azure, SSMS												
4	Milestone 1 - Resource Gathering	1/21/2022	1/28/2022	7		SQL, ML, Power Platform, MS Azure												
5	Milestone 2 - Data Collection/ DB Build	1/25/2022	1/28/2022	3		SQL, Power Platform, MS Azure, ADF												
6	Milestone 3 - Business Rule Review	2/1/2022	2/18/2022	17														
7	Milestone 4 - Develop Functional Testing Requirements	1/17/2022	2/4/2022	18														
8	Phase 2 - Data Engineering	1/17/2022	1/28/2022	11		PowerApps, Power Automate, ADF, SQL, DataVerse, Excel												
9	Milestone 5 - Create Data Connections	1/17/2022	1/22/2022	5		PowerApps, Power Automate, ADF												
10	Milestone 6 - Create Datasets	1/17/2022	1/28/2022	11		SQL, DataVerse, Excel												
11	Milestone 7 - Create Flows/Business Logic	1/17/2022	2/4/2022	18		Power Automate, ADF, PowerApps												
12	Phase 3 - Data Analytics	1/28/2022	2/7/2022	10		PowerBI, PowerApps, SQL												
13	Milestone 8 - Create Visualizations	02/07/2022	2/13/2022	0		PowerBI, PowerApps												
14	Milestone 9 - Create Dashboards	02/07/2022	2/14/2022	0		PowerBI, PowerApps												
15	Milestone 10 - Create Power Apps	02/07/2022	2/15/2022	0		PowerApps, SQL, MS Azure, PowerBI												
16	Phase 4 - Data Science	2/3/2022	2/25/2022	22		Removed From Scope												
17	Milestone 11 - ML Process	Bumped	Bumped	0														
18	Milestone 12 - ML Flow	Bumped	Bumped	0														
19	Milestone 13 - AI Build (Bot/KB)	Bumped	Bumped	0														
20	Phase 5 - Application Development	1/13/2022	2/24/2022	42														
21	Milestone 14 - Design Portal	1/13/2022	2/25/2022	43		Portal Management, HTML, PowerApps												
22	Milestone 15 - Compiling, Linking, Assimilation	1/13/2022	2/26/2022	44														
23	Milestone 16 - Functional Testing	2/20/2022	3/10/2022	18		GitHub												
24	Phase 6 - Testing & Acceptance	2/25/2022	3/10/2022	13														
25	Milestone 17 - UAT	3/4/2022	3/8/2022	4		Functionality Testing -90%												
26	Milestone 18 - Presentation/Acceptance	3/8/2022	3/10/2022	2														
27																		
28																		
29																		
30																		
31																		
32																		

Project Milestones Requirements Rules Issues_Constraints Services And Cost Technician Revenue Factors Technician Revenue Factors (2) Technician Revenue Factors (3) +

Calculation Mode: Automatic Workbook Statistics Give Feedback to Microsoft



Progress Report

Description	Start Date	End Date	Duration	Comp %	KLO
Project Start - TIPA Capstone	1/13/2022	3/10/2022	56	100%	Data Science Immersive
Phase 1 - Technical Feasibility Assessment	1/13/2022	2/4/2022	22	100%	SQL, DataVerse, Excel, MS Azure, SSMS
Phase 2 - Data Engineering	1/17/2022	1/28/2022	11	100%	PowerApps, Power Automate, ADF, SQL, DataVerse, Excel, SSMS
Phase 3 - Data Analytics	1/28/2022	2/7/2022	10	100%	PowerBi, PowerApps, SQL
Phase 5 - Application Development	1/13/2022	2/24/2022	42	100%	Power Portal/Portal Management/ Azure Active Directory
Phase 6 - Testing & Acceptance	2/25/2022	3/10/2022	13	100%	Functionality Testing of Invoices Generating



Azure Resource Administration

Search resources, services, and docs (G+)

David.Clerk@divergenc...
DIVERGENCE ONE (DIVERGENCE...)

Azure services

Create a resource

Cost Management ...

Azure Active Directory

All resources

Azure Network Function...

Monitor

Subscriptions

Resource groups

Data factories

All services

Recent resources

Name	Type	Last Viewed
DB_Email	Logic app	20 hours ago
DC_West	Resource group	21 hours ago
divergencesqldsi202110	SQL server	4 days ago
dcdsi0122db	SQL database	a week ago
dcdsi0122sv	SQL server	a week ago
Azure subscription 1	Subscription	a week ago
dfAzureDataVerse	Data factory (V2)	a week ago
DC_East	Resource group	2 weeks ago
dcAzureGW	On-premises data gateway	3 weeks ago

See all

Navigate





SQL DTL/ELT/ETL via SSMS

The screenshot displays the Microsoft SQL Server Enterprise Manager (SSMS) interface. The left pane shows the Object Explorer with the database structure expanded. The central pane shows a SQL query in the Query Editor, and the bottom pane shows the results of the query in the Results pane.

Query:

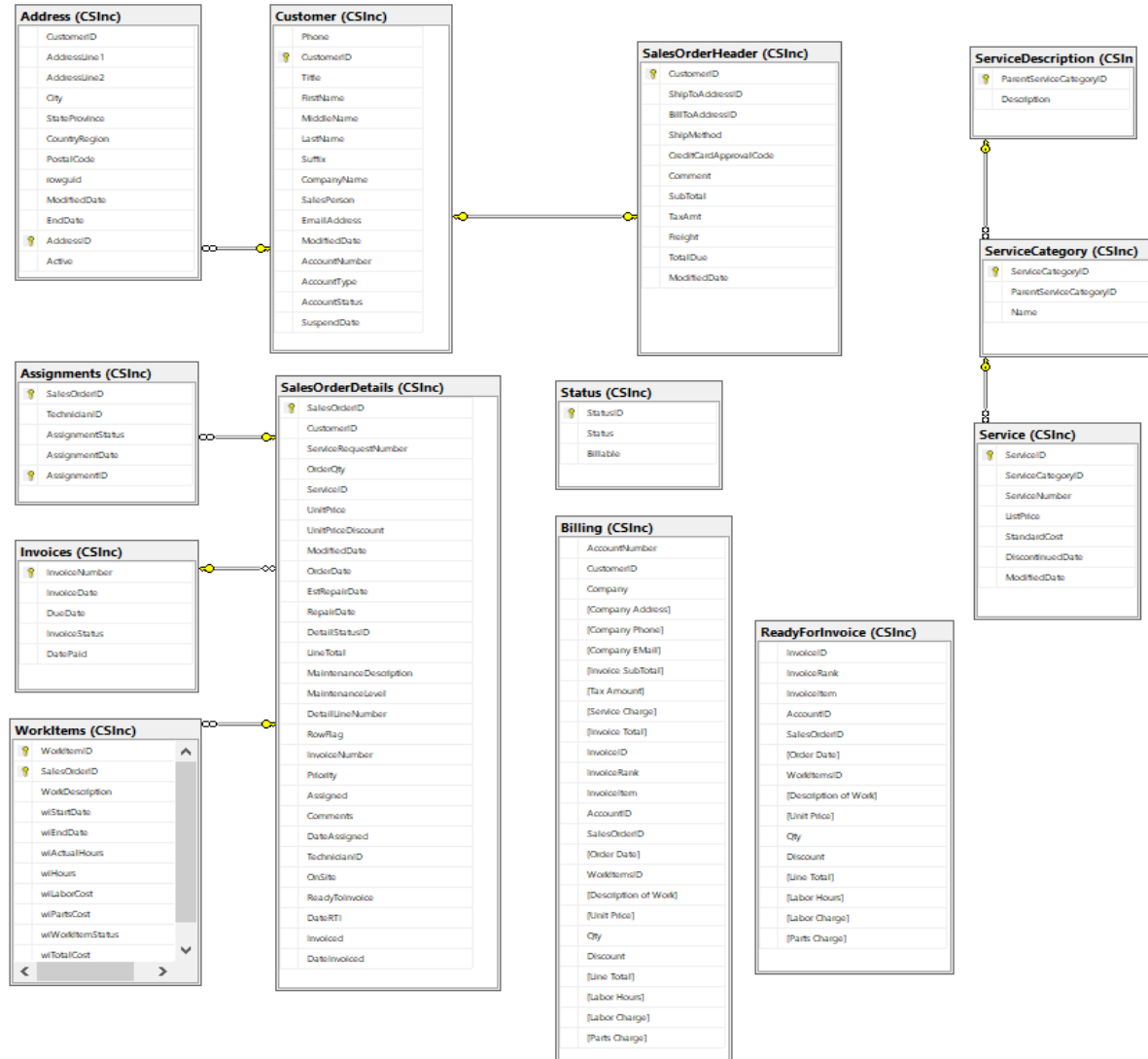
```
SELECT [Object Type], Description, Count(objectID) as Count FROM
(SELECT
    type AS [Object Type], type_desc AS Description, object_id AS objectID
FROM
    sys.objects
WHERE
    (type = 'v' OR
     type = 'p' OR
     type = 'u' OR
     type = 'fn')
    and Name Not like 'sp%' and Name Not like 'usp%'
UNION
SELECT 'D' as type, 'SYS DIAGRAM' as Description, diagram_id as ObjectID FROM sysdiagrams
) as x
GROUP BY [Object Type], Description
ORDER BY [Object Type]
```

Results:

	Object Type	Description	Count
1	D	SYS DIAGRAM	1
2	FN	SQL_SCALAR_FUNCTION	3
3	P	SQL_STORED_PROCEDURE	7
4	U	USER_TABLE	40
5	V	VIEW	25



Database Design





M365 DataVerse/Dynamics Integration

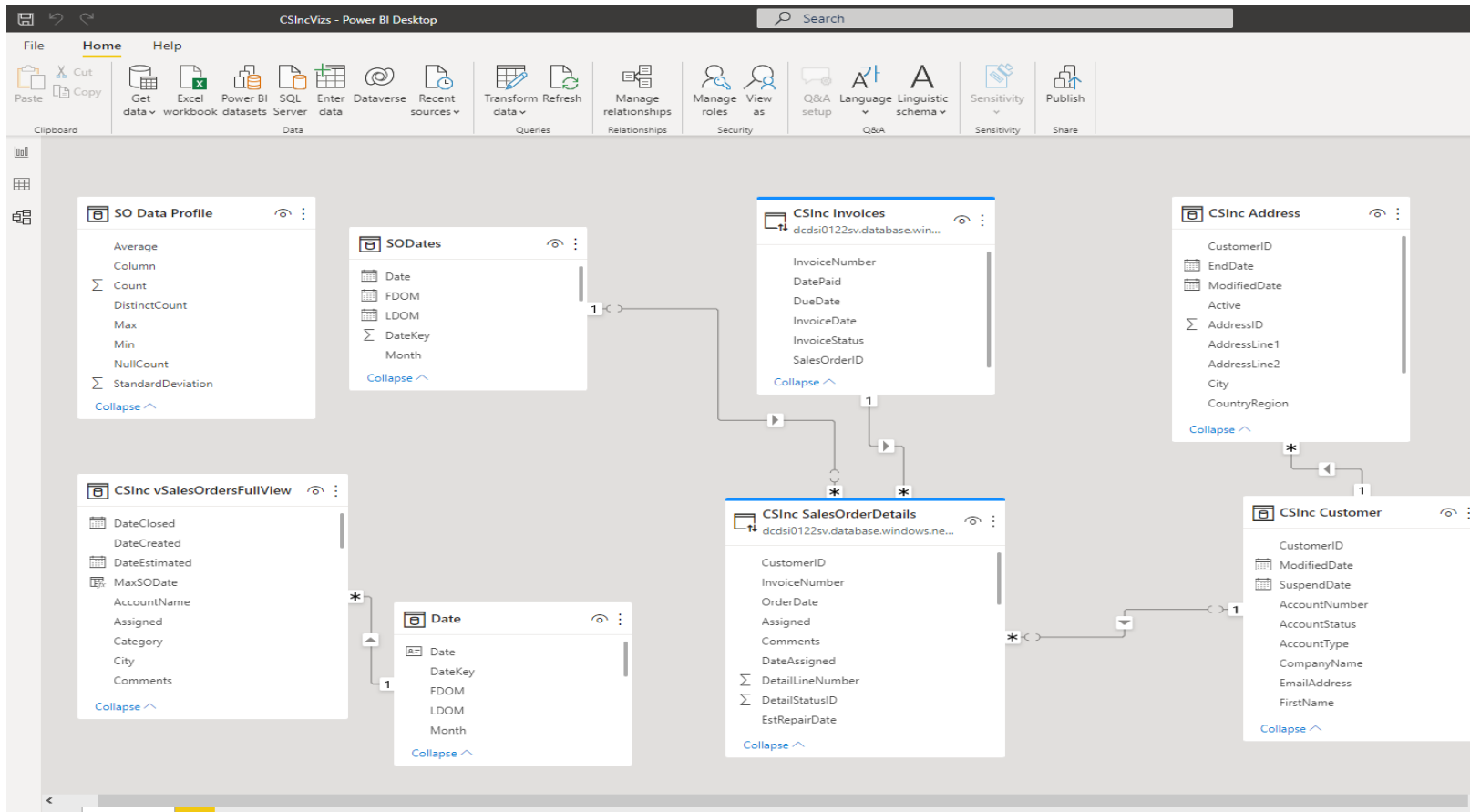
The screenshot shows the Microsoft Power Apps interface. The top navigation bar includes the Power Apps logo, a search bar, and the user's name 'David Clerk'. The left sidebar contains navigation options: Home, Learn, Apps, Create, Dataverse, Tables (selected), Choices, Dataflows, Azure Synapse Link, Connections, Custom Connectors, Gateways, Flows, Chatbots, AI Builder, and Solutions. The main area displays a list of tables under the heading 'Tables'. The table list has columns for Name, Type, Customizable, and Tags. The tables listed are all of type 'Custom' and are customizable.

Table	Name	Type	Customizable	Tags
cr71c_CSIncAssignments	cr71c_csincassignments	Custom	✓	Custom
cr71c_CSIncInvoices	cr71c_csincinvoices	Custom	✓	Custom
cr71c_CSIncPriorityCounts	cr71c_csincprioritycounts	Custom	✓	Custom
cr71c_CSIncProductsAndDescriptions	cr71c_csincproductsanddescriptions	Custom	✓	Custom
cr71c_CSIncReadyToInvoice	cr71c_csincreadytoinvoice	Custom	✓	Custom
cr71c_CSIncServiceCategoryListPrice	cr71c_csincservicecategorylistprice	Custom	✓	Custom
CSInc SalesOrderDetails	cr71c_csincsalesorderdetails	Custom	✓	Custom
CSInc Service	cr71c_csincservice	Custom	✓	Custom
CSInc ServiceCategory	cr71c_csincservicecategory	Custom	✓	Custom
CSInc Technicians	cr71c_csinctechnicians	Custom	✓	Custom
CSInc vSalesOrdersFullView	cr71c_csincvsalesordersfullview	Custom	✓	Custom
Sub-Contractor Registration	csitipa_subcontractorregistration	Custom	✓	Custom
TechnicianApplication	csitipa_technicianapplication	Custom	✓	Custom

Don't see the items you're looking for? Reset the filter above to see more.

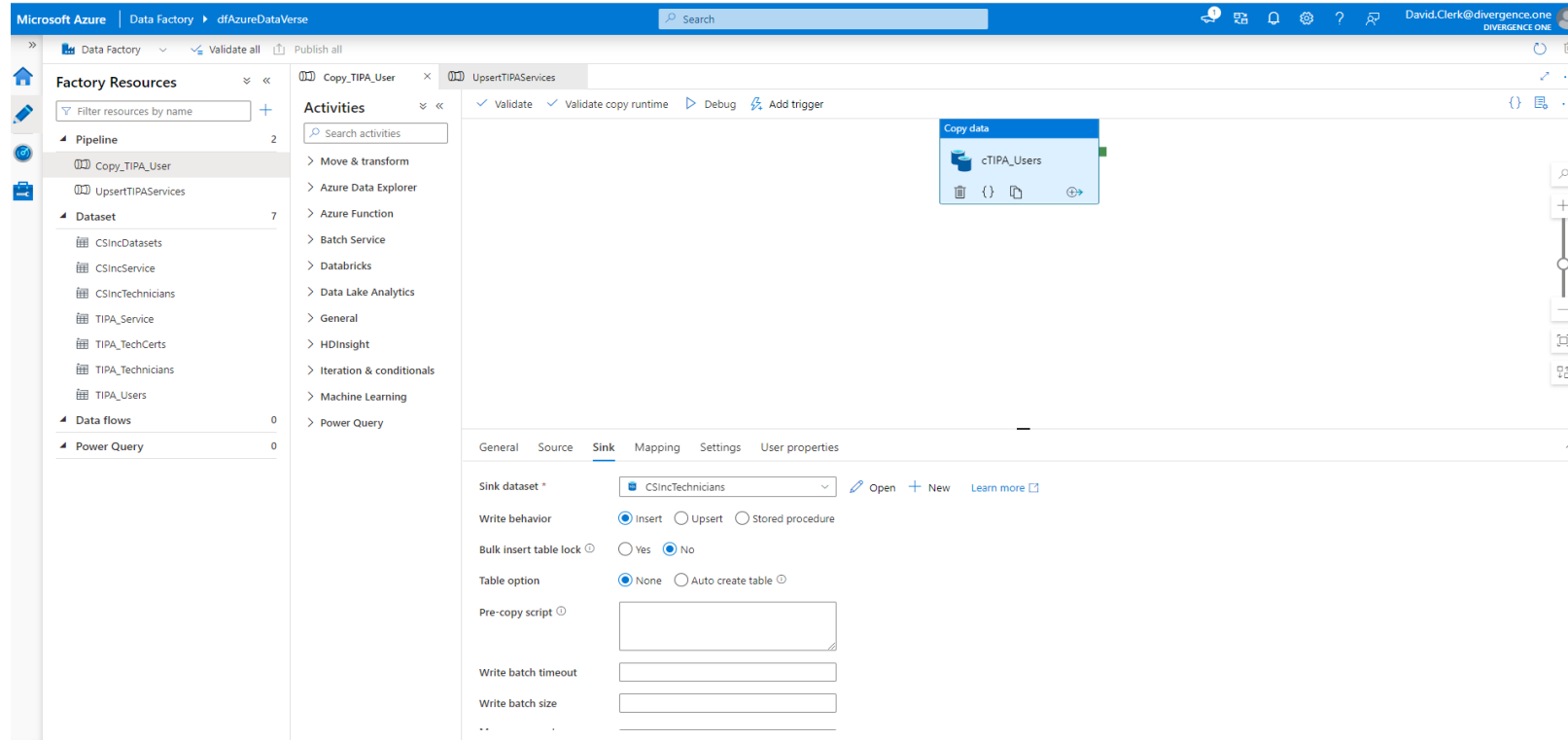


Analytical Datasets - Power BI (Power Query/DAX)





Data Flows - Azure Data Factor





Automation - Power Automate

Power Automate

Search for helpful resources

Environments
David Clerk's Environm...

David Clerk

dcTechOnsiteFI

Comments Save Flow checker Te

PowerApps

Execute stored procedure (V2)

+ New step Save

```
/****** Object: StoredProcedure [CSinc].[Invoicing] Script Date: 2/17/2022 6:18:33 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
-- Create Stored Procedure Template for Azure SQL Database
-- *****
-- Author: David Clerk
-- Create Date: 02/16/2022
-- Description: Create Datasets For Invoice
-- *****
ALTER PROCEDURE [CSinc].[Invoicing]
AS
BEGIN
    SET NOCOUNT ON
    DECLARE @Acct as NVarchar(100)
    DECLARE @Sold as int, @Invid as int
    -- TRUNCATE TABLE CSinc.ReadyForInvoice
    INSERT INTO CSinc.ReadyForInvoice
    SELECT Null, * FROM
    (SELECT
        k.InvoiceRank, k.InvoiceItem, k.AccountID, k.SalesOrderID, k.OrderDate, k.WorkItemsID, k.[Description of Work], k.[Unit Price], k.Qty, k.Discount, k.[Line Total], k.[Labor Hours], k.[Labor Charge], k.[Parts Charge]
    FROM
        CSinc.ReadyForInvoice AS k LEFT OUTER JOIN
        CSinc.ReadyForInvoice AS m ON k.SalesOrderID = m.SalesOrderID
    WHERE
        (m.SalesOrderID IS NULL)
    GROUP BY k.InvoiceRank, k.InvoiceItem, k.AccountID, k.SalesOrderID, k.OrderDate, k.WorkItemsID, k.[Description of Work], k.[Unit Price], k.Qty, k.Discount, k.[Line Total], k.[Labor Hours], k.[Labor Charge], k.[Parts Charge]) as x2
    DECLARE Iv_cursor CURSOR FOR
    SELECT AccountID, [SalesOrderID]
    FROM [CSinc].[ReadyForInvoice]
    OPEN Iv_cursor
    FETCH NEXT FROM Iv_cursor INTO @Acct, @Sold
    WHILE (@FETCH_STATUS = 0)
    BEGIN
        INSERT INTO [CSinc].[Invoices] (InvoiceDate, InvoiceStatus)
        Values (GetDate(), 'Invoiced')
        SELECT @Invid = MAX(InvoiceNumber) FROM [CSinc].[Invoices]
        UPDATE [CSinc].[ReadyForInvoice]
        SET InvoiceID = @Invid
        WHERE AccountID = @Acct
        UPDATE [CSinc].[SalesOrderDetails]
        SET InvoiceNumber = @Invid
        WHERE SalesOrderID = @Sold
        UPDATE [CSinc].[SalesOrderDetails]
        SET DateInvoiced = GetDate()
        WHERE SalesOrderID = @Sold
        UPDATE [CSinc].[SalesOrderDetails]
        SET Invoiced = 1
        WHERE SalesOrderID = @Sold
        FETCH NEXT FROM Iv_cursor INTO @Acct, @Sold
    END
    INSERT INTO CSinc.Billing Select * FROM CSinc.Unbilled
    CLOSE Iv_cursor;
    DEALLOCATE Iv_cursor;
END
```



Web - Power Pages

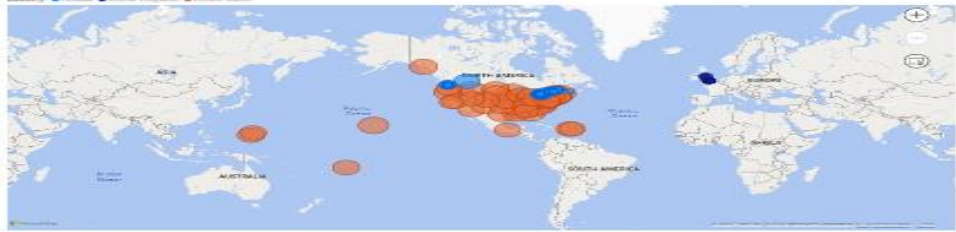
CSI Inc. Home Technicians - CSInc Admin - Contact us Sign in

Don't Be Gray... Make More Green With CSInc!



Sign-in and Apply Now - To become one of our Freelance Technicians.
Check For Locations Near You!

CSInc Locations



Filters

Search

Filters on all pages

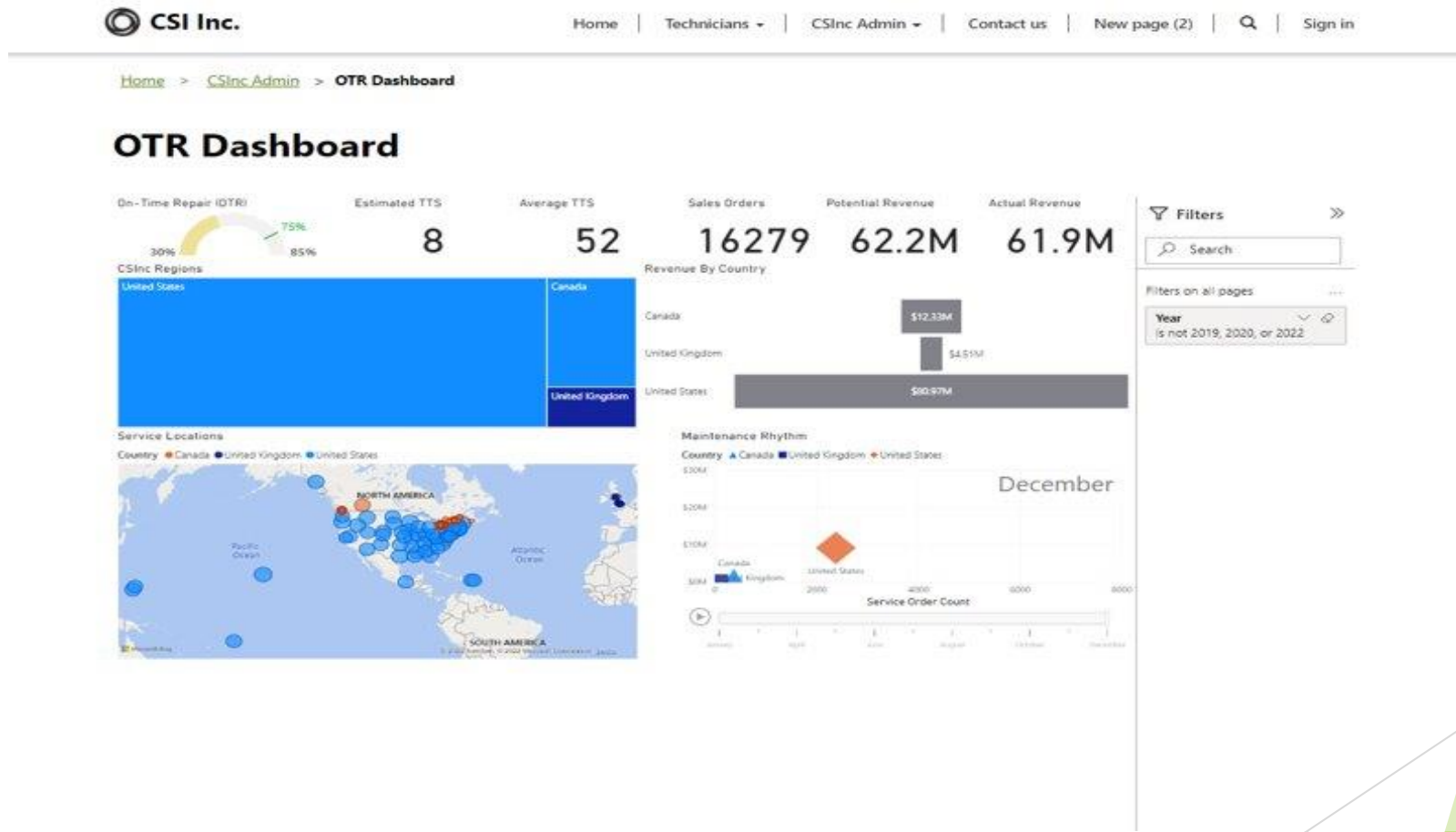
Year

Is not 2019, 2020, or 2021



Portal Integrations

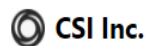
Power Bi/Power Apps in Power Pages





Dashboard

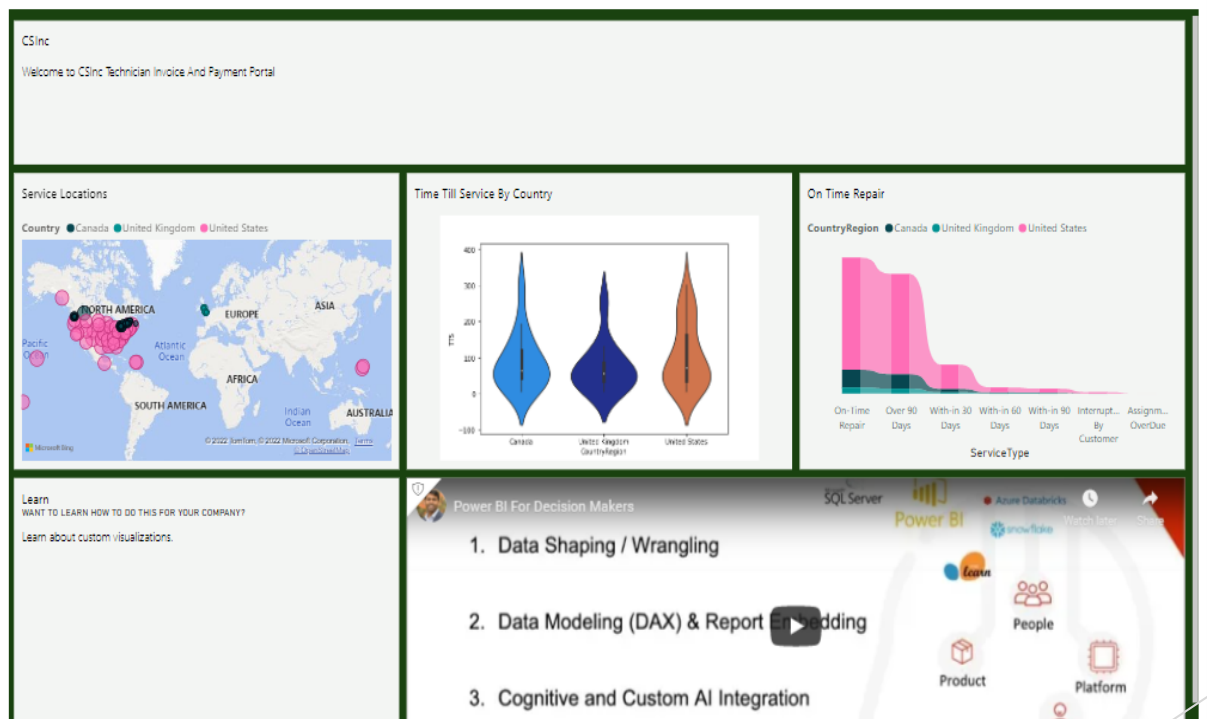
Power Bi w/Customized Python Visualization Embedded In Power Pages



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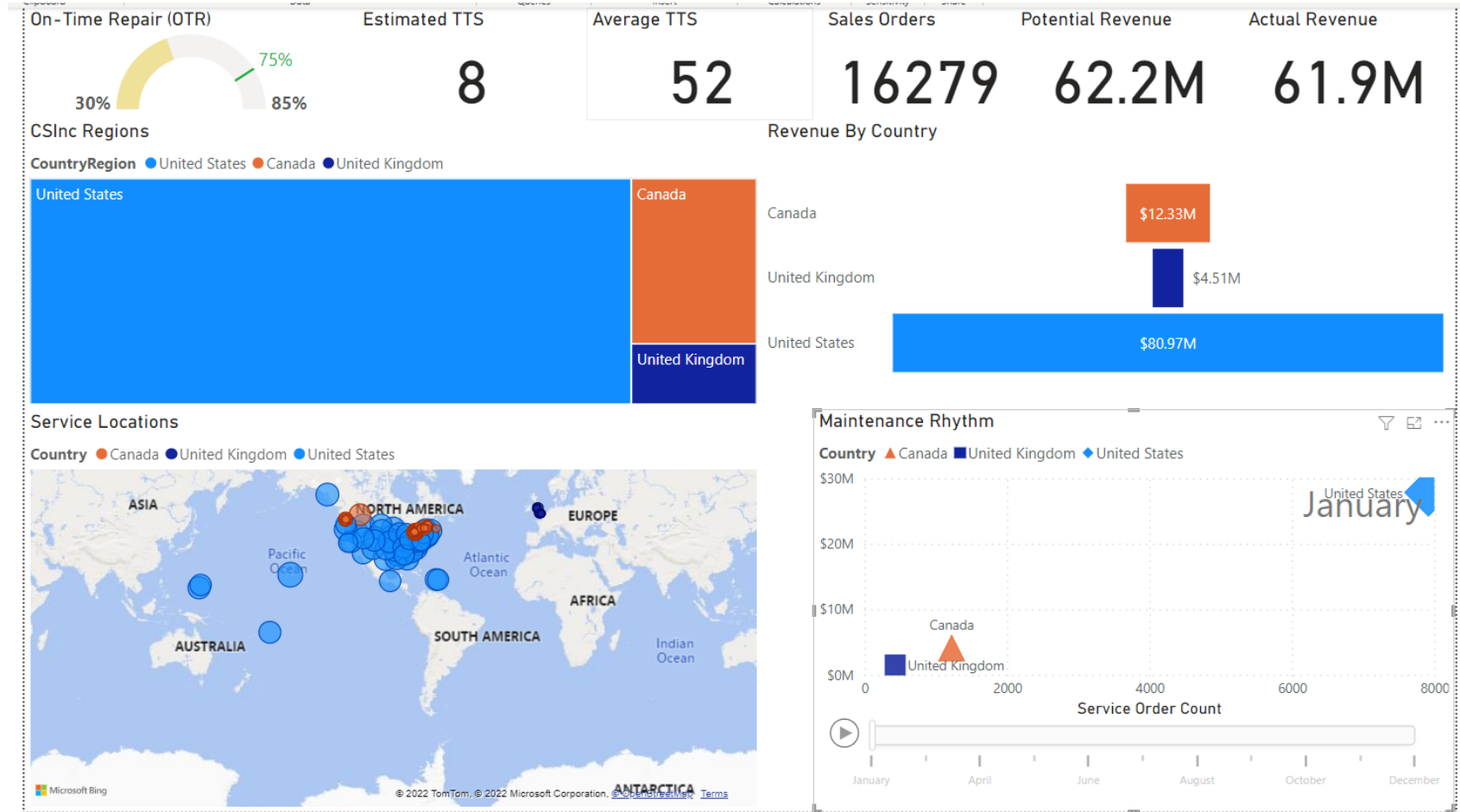
[Home](#) > [CSInc Admin](#) > [OTR Dashboard](#)

OTR Dashboard



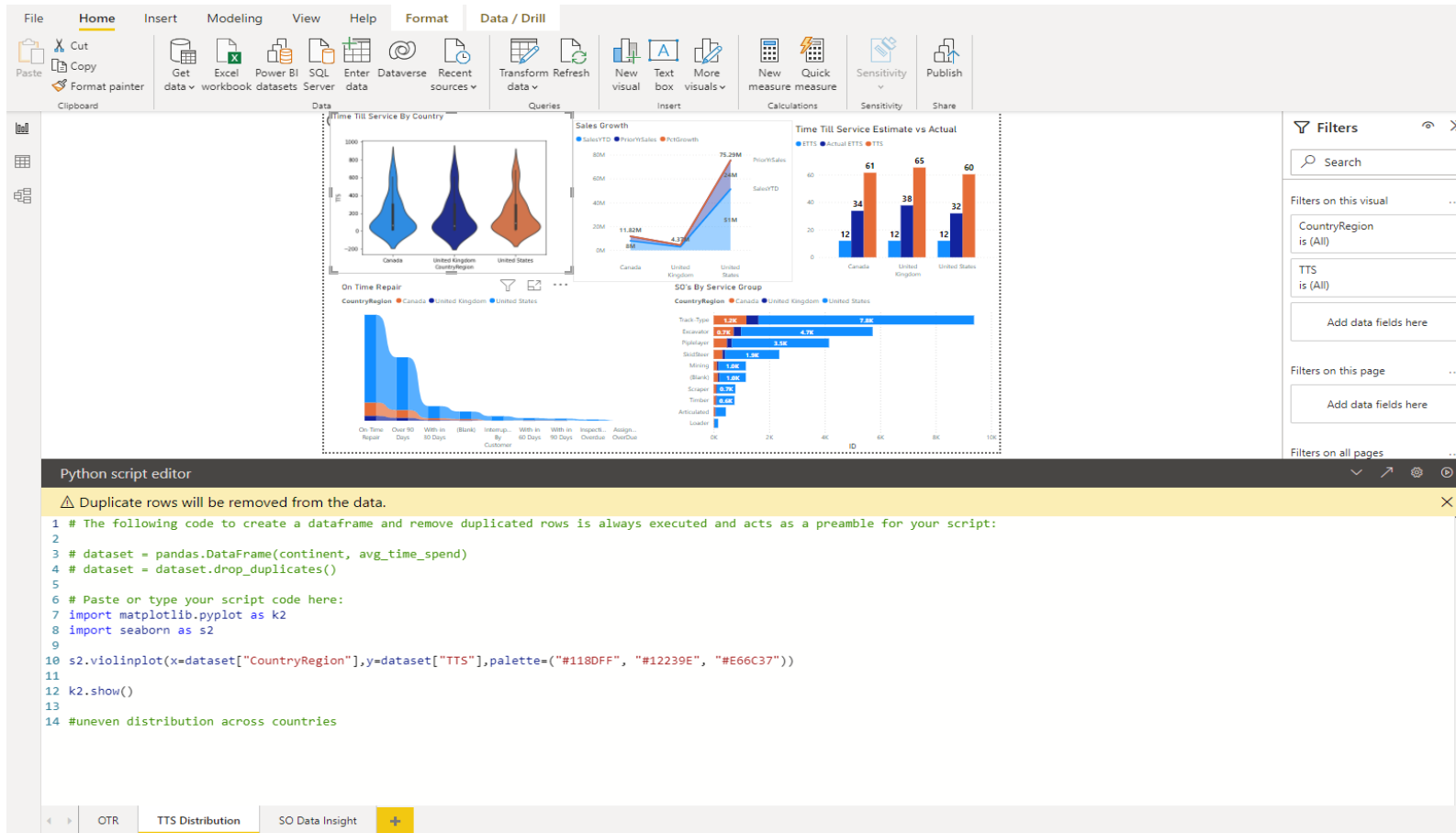


KPIs, Metrics, and Animations





Python Integration





Customized Apps

Power Apps | Canvas Environment: David Clerk's Environm...

Service Order Tasks

100
All SOs

0
New SOs

100
Repairs In-Progress

100
Complete

0
Inspection In-Progress

0
Pending Parts/Tool/Supplies

0
On-Hold (Customer Request)

0
Test

Welcome
David.Clerk@divergence.one

Search By SO Number: Search By SO Num

Search By Company: Search By Company Name

Filter By SO Status: Repairs In-Progress

SO Num: 8800100 Priority: Normal Company: A Machine Store 8713 Yosemite Ct. Montgomery, Alabama, 36104 orlando0@adventure-works.com 367-555-0246 Estimated Repair Cost: \$372.00 Technician: david.clerk@divergence.one	Date Assigned: 2/15/2022 Assignment Status: Assigned	SO Date: 09/08/2021 ESD: 09/18/2021 SO Status: Repairs In-Progress Completion Date: Maintenance Description: PMCS Service Category/Type: Excavator - Power Shovel Maintenance Level: 1	Send En Route Notification Check-In/Out As 'On-Site' Manage Work Items List	Work Items List 8800100 33374 Perform 1000Hr/Annual PMCS IAW Technical Manual - Parts Kit Provided. 8800100 33378 Replace Tires 8800100 33382 Replace Accumulator Seals 8800100 33386 Bleed And Purge Pneumatic Braking System 8800100 33390 Replace Winch Cable 8800100 33394 Replace Roosa Master
SO Num: 8800101 Priority: Normal Company: Progressive Equipment 1318 Lasalle Street Juneau, Alaska, 99801 keith0@adventure-works.com 645-555-0333 Estimated Repair Cost: \$1002.00 Technician: david.clerk@divergence.one	Date Assigned: 2/15/2022 Assignment Status: Assigned	SO Date: 09/13/2021 ESD: 09/18/2021 SO Status: Repairs In-Progress Completion Date: 02/16/2022 Maintenance Description: PMCS Service Category/Type: Mining - Construction & Mining Trucks Maintenance Level: 1	Send En Route Notification Check-In/Out As 'On-Site' Manage Work Items List	



Service Order Tasks

100010000000

All SOsNew SOsRepairs In-ProgressCompleteInspection In-ProgressPending Parts/Tool/SuppliesOn-Hold (Customer Request)Test

Welcome

Search By SO Number:

Search By SO Num

Search By Company:

Search By Company Name

Filter By SO Status:

Repairs In-Progress

SO Num: 8800100

Priority: Normal

Company:
A Machine Store
8713 Yosemite Ct. Montgomery, Alabama, 36104
orlando0@adventure-works.com
367-555-0246

Estimated Repair Cost: \$372.00

Technician: david.clerk@divergence.one

Date Assigned: 2/15/2022

Assignment Status: Assigned

SO Date: 09/08/2021

ESD: 09/18/2021

SO Status: Repairs In-Progress

Completion Date:

Maintenance Description: PMCS

Service Category/Type: Excavator - Power Shovel

Maintenance Level: 1

Send En Route Notification

Check-In/Out As 'On-Site'

Manage Work Items List

SO Num: 8800101

Priority: Normal

Company:
Progressive Equipment
1318 Lasalle Street Juneau, Alaska, 99801
keith0@adventure-works.com
645-555-0333

Estimated Repair Cost: \$1002.00

Technician: david.clerk@divergence.one

Date Assigned: 2/15/2022

Assignment Status: Assigned

SO Date: 09/13/2021

ESD: 09/18/2021

SO Status: Repairs In-Progress

Completion Date:
02/16/2022

Maintenance Description: PMCS

Service Category/Type: Mining - Construction & Mining Trucks

Maintenance Level: 1

Send En Route Notification

Check-In/Out As 'On-Site'

Manage Work Items List

Work Items List

8800100

33374 Perform 1000Hr/Annual PMCS IAW Technical Manual - Parts Kit Provided.

8800100

33378 Replace Tires

8800100

33382 Replace Accumulator Seals

8800100

33386 Bleed And Purge Pneumatic Braking System

8800100

33390 Replace Winch Cable

8800100

33394 Replace Roosa Master



Customized Apps

Power Apps | Canvas

Environment
David Clerk's Environm...

Work Items List

SO Num	Work Item Description	Status	Start
8800100	Perform 1000Hr/Annual PMCS...	In-Progress	9/8/2021 1
8800100	Replace Tires	In-Progress	9/8/2021 1
8800100	Replace Accumulator Seals	In-Progress	9/8/2021 1
8800100	Bleed And Purge Pneumatic Br...	In-Progress	9/8/2021 1
8800100	Replace Winch Cable	In-Progress	9/8/2021 1
8800100	Replace Roosa Master	In-Progress	9/8/2021 1
8800100	Repair ROPS Seal	In-Progress	9/8/2021 1

***SalesOrderID**
8800100

Work Item Description
Perform 1000Hr/Annual PMCS IAW Technical Manual - Parts Kit Provided.

Start Date
9/8/2021

Date Complete
12/31/2001

wiHours
8.5

wiActualHours
8

wiWorkItemStatus
In-Progress

wiLaborCost
0

wiPartsCost
0

Submit

WorkItemID
22274



TIPATechnician Invoice Payment Application

123 Academy Drive
Texas
USA 76028
csinctipa@divergence.fun

John Doe
Contoso Ltd.
55 East 52nd Street 21st
New York
US 10022

INVOICE 1

10/02/2017
Payment Terms: 30 days

#	SaleOrderID	Order Date	WorkItemID	Description	Qty/Hrs	Unit Price	Line Total
0	Product Name #1	Product Name #1	Product Name #1		1	499	499
1	Product Name #2	Product Name #2	Product Name #2		1	599	599
			\$ Invoice Total	\$1098			

Payment Details

Divergence One
Bank/Sort Code: 32-75-97
Account Number: 28270761
IBAN: 973547
BIC: 220197
Payment Reference: INV 1

Other Information

Payment should be made by bank transfer or cheque made payable to CSInc TIPAL

Questions?

A more detailed Project Proposal is
available [@DC_CapstoneProject](#)

